Reviewer 2: Anonymous

Review remarks on the manuscript entitled "**YIGSR sequence contained elastin like recombinamer for sustaining retinal pigment epithelium as a possible tool for retinal regenerative medicine".**

In the paper the authors have provided a method for sustaining retinal pigment (RPE) using elastin like recombinamer (ELR) that contains a specific YIGSR motif for enhanced cell binding, proliferation and spreading of various cell-types. Using a human RPE cell line, ARPE19, the authors have demonstrated that elastin like recombinamer having YIGSR motif can support RPE cell growth and viability. ELR as synthetic biopolymers have been used increasingly in the field of cancer therapy and tissue scaffolding. Using a laminin sequence present in RPE basement membrane that promotes RPE cell growth in conjunction to the ELR scaffold is promising. Moreover, expression of Ecad-1 and ZO-1 proteins, and presence of tight and gap junctions between growing ARPE19 cells is suggestive of the fact that ELR-YIGSR substrate is suitable to sustain epithelial nature of ARPE19 cells This can provide a new approach towards RPE cell replacement therapy in age-related macular degeneration. The manuscript is clearly written and the research is well-performed.

Some improvement could however be provided although not of major concern: 1) the growth and number of ARPE19 cells in ELR-YIGSR scaffold is too low than that of TCP polystyrene control. There might be growth inhibitory factors induced by ELR-YIGSR layer that needs to be reviewed and addressed by the authors.

2) There are other minor issues in the manuscript that requires correction from the authors. Firstly, there are many typological and grammatical errors which change the meaning of entire sentence. I would like to draw attention of the authors to line no. 390 and 391 and would suggest to make the necessary changes. There are several similar errors in the entire manuscript which requires corrections. The idea proposed by the authors using ELR-YIGSR as a polymeric scaffold to support growth of RPE cells is done only in a cell line and needs further testing with primary human cells. Since, the authors too have mentioned the need for the same it can be addressed in subsequent studies.